

Transistors, processors, circuits, interfaces, cables and peripherals: These are essentially what computers and computing have been about for the last few decades. Until recently, electricity and magnetism have played a major role in all of these. Even now, fibre-optic cables, optical storage devices such as CD-ROMs, and optical mice are probably the only aspects of networking and computing where you're likely to see optics at work.

This scenario is set to change. Light will play more and more of a role in networks and computing systems. However, calling optical computing 'the next wave of comput-

ing' is not new. It was called the next wave of computing back in the '80s, but except for optical storage and optical fibres, nothing much came out of the research labs, until recently.

But why light? Why would we want computers to be optical, instead of electronic, or silicon-based?

Simply put, the need for computing at the speed of light comes from the need for speed. Silicon-based circuits will soon reach miniaturisation limits: Experts say this will be sometime around 2030, when such computing circuits will hit a fundamental physical roadblock on the path to miniaturisation. Incidentally, miniaturisation has been the key to

increased speeds.

Optics is probably one of the ways out. However, it's not a new field—in fact, it's almost ancient—and neither is its application to computing anything new. However, research in optics—especially non-linear optics—with computing as the focus has never been pursued with the kind of vigour we're seeing today.

It's said that the 20th century was the century of electronics, and the 21st will be that of optics.

Here, we take a look at what optoelectronic and optical artefacts have been churned out of the labs so far, and what this means for fully-optical computing.

Computing at Lightspeed!

Light is all set to make inroads a computer... in the not-too-distant tomorrow

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